

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Experiment 12: Iris Flower Classification using K-Nearest Neighbours (KNN)

Aim

To implement the K-Nearest Neighbours (KNN) algorithm in Python to classify Iris flowers into different species.

Procedure

1. Import the required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Standardize the feature values.
5. Create the KNN classifier with $K = 3$.
6. Train the classifier using the training data.
7. Predict the classes of the test data.
8. Calculate the accuracy and display the confusion matrix.
9. Predict the class of a new Iris flower.

Algorithm

1. Start.
2. Load the Iris dataset.
3. Separate the features and target classes.
4. Split the data into training and testing sets.
5. Standardize the features.
6. Select $K = 3$.
7. Calculate the distance between the test sample and training samples.
8. Select the 3 nearest neighbours.

9. Assign the majority class to the test sample.
10. Calculate the classification accuracy.
11. Classify a new Iris flower.
12. Stop.

Dataset 1

Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
5.1	3.5	1.4	0.2	Setosa
4.9	3.0	1.4	0.2	Setosa
6.0	2.9	4.5	1.5	Versicolor
5.7	2.8	4.1	1.3	Versicolor
6.5	3.0	5.5	1.8	Virginica

Dataset 2

Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
5.0	3.4	1.5	0.2	Setosa
5.2	3.5	1.5	0.2	Setosa
5.9	3.0	4.2	1.5	Versicolor
6.1	2.8	4.7	1.2	Versicolor
6.7	3.1	5.6	2.4	Virginica

Dataset 3

Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
5.1	3.8	1.5	0.3	Setosa
4.7	3.2	1.3	0.2	Setosa
6.2	2.2	4.5	1.5	Versicolor
5.6	2.9	3.6	1.3	Versicolor
6.9	3.1	5.4	2.1	Virginica

Dataset 4

Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
4.8	3.1	1.6	0.2	Setosa
5.4	3.7	1.5	0.2	Setosa
5.8	2.7	4.1	1.0	Versicolor
6.0	2.7	5.1	1.6	Versicolor
6.4	2.8	5.6	2.1	Virginica

Dataset 5

Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
5.0	3.6	1.4	0.2	Setosa
5.3	3.7	1.5	0.2	Setosa
5.5	2.4	3.8	1.1	Versicolor
6.0	2.9	4.5	1.5	Versicolor
6.5	3.0	5.2	2.0	Virginica

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

data = load_iris()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = KNeighborsClassifier(n_neighbors=3)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Labels:", y_test)
print("Predicted Labels:", y_pred)

print("Accuracy:", accuracy_score(y_test, y_pred))

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

new_flower = [[5.1, 3.5, 1.4, 0.2]]

new_flower = scaler.transform(new_flower)

prediction = model.predict(new_flower)

print("New Flower Classification:", data.target_names[prediction[0]])
```

Output

```
*** Actual Labels: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Predicted Labels: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Accuracy: 1.0
Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
New Flower Classification: setosa
```

Result

The Iris Flower Classification program was successfully implemented using the K-Nearest Neighbours algorithm. The model classified Iris flowers into Setosa, Versicolor, and Virginica species with high accuracy.